

Yixiao Wang

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SUMMARY

Incoming PhD student in Computer Science at Duke University with 1+ years of machine learning research experience. Strong background in scalable algorithms, optimization, deep learning, statistical modeling, and discrete optimization. Skilled at turning mathematical insights into efficient, production-ready implementations in C++ and Python. First-author publications at NeurIPS, ICML, and ICLR, with additional conference and workshop papers.

EDUCATION

- **Duke University** Aug. 2024 – Present
M.S. in Statistical Science | Incoming Ph.D. Student in Computer Science GPA: 3.95/4.00
- **University of Science and Technology of China, School of the Gifted Young** Sep. 2020 – Jul. 2024
B.S. in Mathematics Top 10%, Outstanding Graduate

RESEARCH EXPERIENCE

- **Duke University, Interpretable Machine Learning Lab** Jun. 2025 – Present
Supervisor: [Prof. Cynthia Rudin](#) Durham, NC, United States
 - **Developed near-optimal algorithms for sparse regression trees in seconds**, unifying constant-leaf and linear-leaf formulations under a dynamic-programming lookahead framework.
 - **Established computational complexity bounds and proved approximation guarantees**, showing that the proposed tree-optimization algorithms achieve solutions provably close to the global optimum.
 - **Built an optimized C++/Python package** with reproducible benchmarks on 10 UCI datasets, demonstrating consistent order-of-magnitude speedups over optimal baselines (OSRT, STreeD) while preserving accuracy.
- **Duke University, Department of Computer Science** Jun. 2025 – Present
Supervisor: [Prof. Anru Zhang](#) Durham, NC, United States
 - **Co-first-author, NeurIPS 2025 Workshop on What Can(t) Transformers Do? (Oral)**: Established the first rigorous in-context learning theory for time series forecasting, showing that Transformers are fundamentally suboptimal: proved a strictly positive excess-risk gap over linear predictors, quantified it with an inverse-linear convergence rate, and demonstrated exponential error compounding under Chain-of-Thought rollout.
 - **Co-first-author, ICLR 2026**: Modeled hidden-state trajectories of Transformers using Menger curvature to analyze logical consistency, uncovering geometric invariants that govern reasoning behavior.
- **Duke Center for Computational Evolutionary Intelligence (CEI)** Jan. 2025 – Present
Supervisor: [Prof. Yiran Chen](#) Durham, NC, United States
 - **Developed ZEUS**, a controllable training-free diffusion acceleration framework that achieves up to $3.2\times$ **end-to-end speedups** ($2.0\times$ – $3.2\times$ across SD-2, SDXL, and Flux) while improving perceptual fidelity ($\text{LPIPS} \leq 0.08$, $\text{FID} \leq 3.5$) over prior training-free methods.
 - **Co-first-author, ICML 2025**: Developed **SADA**, a stability-guided, training-free adaptive diffusion accelerator that achieves $\geq 1.8\times$ generation speedups across SD-2, SDXL, and Flux, while maintaining perceptual quality ($\text{LPIPS} \leq 0.10$, $\text{FID} \leq 4.5$).
 - **Contributing author, AAI 2026**: Provided theoretical analysis for **FlashSVD**, deriving memory–compute complexity bounds and characterizing rank-dependent speedup and compression thresholds for low-rank Transformer inference.
- **Wake Forest University, Department of Computer Science** Jan. 2025 – Aug. 2025
Supervisor: [Prof. Aditya Devarakonda](#) Winston-Salem, NC, United States
 - **Co-first-author NeurIPS 2025**: Proposed and led the development of **Enhanced Cyclic Coordinate Descent (ECCD)**, a Taylor-corrected block coordinate descent method for generalized linear models. The algorithm reduces repeated link-function evaluations and achieves provable improvements in computational efficiency and error bounds.
 - **Developed a high-performance C++-based R package** with optimized single-point and pathwise λ solvers, supporting large-scale regularized regression with strong numerical stability.
 - **Demonstrated 2 – $4\times$ speedups** over GLMNet while maintaining convergence and accuracy, validated through extensive benchmarks against state-of-the-art R/Python implementations.

PUBLICATIONS AND PREPRINTS

C=CONFERENCE, P=PREPRINT, S=SUBMITTED

- [C.1] **Yixiao Wang***, Zishan Shao*, Ting Jiang, Aditya Devarakonda.
“**Enhanced Cyclic Coordinate Descent Methods for Elastic Net Penalized Linear Models**,” *NeurIPS* 2025.
[Paper] [Code]
- [C.2] Ting Jiang*, **Yixiao Wang***, Hancheng Ye*, Zishan Shao, Jingwei Sun, Jingyang Zhang, Zekai Chen, Jianyi Zhang, Yiran Chen, Hai Li. “**SADA: Stability-guided Adaptive Diffusion Acceleration**,” *ICML* 2025. [Paper]
[Code] [Page] [Slides]

- [C.3] Yufa Zhou*, **Yixiao Wang***, Surbhi Goel, Anru R. Zhang.
"Why Do Transformers Fail to Forecast Time Series In-Context?" *NeurIPS Workshop on What Can('t) Transformers Do?* 2025 (Oral Presentation). [Paper] [Code]
- [C.4] Yufa Zhou*, **Yixiao Wang***, Xunjian Yin*, Shuyan Zhou, Anru R. Zhang.
"The Geometry of Reasoning: Flowing Logics in Representation Space," *ICLR* 2026. [Paper] [Code] [Dataset]
- [C.5] Zishan Shao, **Yixiao Wang**, Qinsi Wang, Ting Jiang, Zhixu Du, Hancheng Ye, Danyang Zhuo, Yiran Chen, Hai Li. **"FlashSVD: Memory-Efficient Inference with Streaming for Low-Rank Models,"** *AAAI* 2026. [Paper] [Code]
- [S.1] **Yixiao Wang***, Hayden McTavish*, Varun Babbar*, Margo Seltzer, Cynthia Rudin. **"CLARITree: Cholesky and Lookahead Accelerations for Regression with Interpretable Piecewise Linear Trees,"**.
- [S.2] **Yixiao Wang***, Ting Jiang*, Zishan Shao*, Hancheng Ye, Jingwei Sun, Mingyuan Ma, Jianyi Zhang, Yiran Chen, Hai Li, **"Accelerating Denoising Generative Models is as Easy as Predicting Second-Order Difference,"**. [Paper][Code] [Page]
- [S.3] Zishan Shao, Lixun Zhang, Kangning Cui, **Yixiao Wang**, Ting Jiang, Hancheng Ye, Qinsi Wang, Zhixu Du, Yuzhe Fu, Fan Yang, Danyang Zhuo, Yiran Chen, Hai Helen Li, **"DecodeShare: Tracing the Shared Pathways of LLM Decode-Time Decisions,"**.

* denotes co-first authorship; first-/co-first-authored works listed first.

HONORS AND AWARDS

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|---|-------------|
| • Centennial & Graduate Professional Fellowship (nominated fellowship, Duke University) | 2026 |
| • Outstanding Graduate, University of Science and Technology of China (USTC) | 2024 |
| • China Petroleum Scholarship (3 awardees, School of the Gifted Young, USTC) | 2023 |
| • Excellent Teaching Assistant (Top 3 in the University, USTC) | 2023 |
| • Silver Prize, Outstanding Student Scholarship (Top 15%, USTC) | 2022 & 2021 |
| • "Qiang Wei Feng Gong De Yu" (Diligence and Moral Conduct) Scholarship (USTC) | 2022 |
| • Excellent President of Student Club (Origami Club, USTC) | 2022 |
| • First Prize, Chinese Mathematical Competition for University Students (Top 1%) | 2022 & 2021 |
| • Bronze Prize, Outstanding Freshmen Scholarship (Top 35%, USTC) | 2020 |

SERVICE

- Teaching Assistant: Linear Algebra B1 (Spring 2023, Top 3 University-wide)
- Reviewer: NeurIPS 2026; AISTATS 2026; ICML 2026; AAAI Workshop 2026
- Research Mentorship
 Ting Jiang (Duke undergraduate, now PhD student at Carnegie Mellon University) Jan 2025 – Sep 2025
- Leadership: President (Origami Club in USTC)
- Volunteer: Hefei Chunyu Parent Support Center, 2021–2022

SELECTED COURSE PROJECTS

- **Airbnb Price Prediction in New York City** (CS 671 Theory & Algorithms of Machine Learning, Duke University) – Achieved top 5 out of 137 participants using advanced stacking strategies, combining XGBoost, LightGBM, and 20+ weak learners for robust generalization. [Report][Code]

ADDITIONAL INFORMATION

Languages: English, Mandarin, Sichuanese dialect.

Programming & Software: Python (primary), C/C++, $\LaTeX$, Linux, HTML/CSS, MATLAB, R.

Interests: Origami and other visual arts (selected works available at [portfolio link](#)).